20.	Field oxidation 1-2# FIELD2	3# Clean Oxidation into furnace Oxidation outof furnace		Thickness of SiO2 （X.val=9.2）
21.	Remove all of SiN （10:1HF 1min/160°C 60min）	Etch SiO2 Etch SiN check	No simulation	
22.	Etch SiO_2 with specified thickness		According to the extracted thickness of SiO2 in step20, etch the right thickness, and remain 800±10nm SiO2 left. X.val=9.2 x.val for this etching is the whole region	Thickness of SiO2 need to be etched 证明：如下例
23.	Gate oxide 2 (30±5nm) 1-3# 350ATEST	RCA Clean Oxidation into furnace Oxidation outof furnace		Thickness of SiO2 X=4.5
24.	Deposit poly	3# Clean CVD into furnace CVD into furnace	350±30nm P_{31} doped 5E12	
25.	Poly Photolithography mask:POLY	Pretreatment Photo deposit Exposure Development Check	Note: Photolithography should be judged by yourself.	
26.	Etch poly	Etch poly	According to the shape of poly to etch poly	
27.	Remove all of photo	Dry etch photo Wet etch photo check	should be judged by yourself.	

例：Step 22, Thickness of SiO2 need to be etched 666.66um

```
EXTRACT> extract name="ox22" thickness material="SiO~2" mat.occno=1
ox22=8000.87 angstroms (0.800087 um)  X.val=8
```

FIELD2 1-2 field oxidation								
Step	Temp °C	Ramp °C/min	Time	N2(slpm)	O2(slpm)	H2(slpm)	Tca (sccm)	Boat Cm/min
Push in	800		30’	20				
Rampto	950		30’	8				
Anneal	950		1h	8				
Tca+dryox	950		15’30”		8			
Wet ox	950		6h		4.0	7.3		
Ramp to	800		50’	8				
Ramp to	750		10’	8				
Push out	750		15’	10				

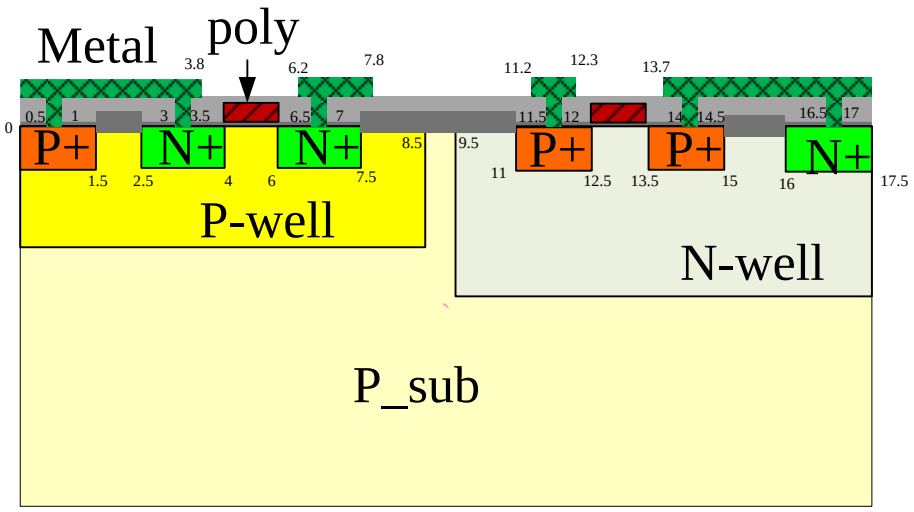
Note: if there are two types of gas in one step, the code is like this: diffus time=30 temp=950 f.a=4 f.b=5

350ATEST 1-3 gate oxide 2

CMOS Process

Step	Temp °C	Ramp °C/min	Time	N2(slpm)	O2(slpm)	H2(slpm)	Tca (sccm)	Boat Cm/min
Push in	750							
Ramp to	1000		20’	3				
Dry O2	1000		33’	Dry oxidation				
Ramp to	750		20’	Use inert				

Note：diffusion time=* temp=* t.final=* inert



NWELL PWELL ACTIVE NPLUS PPLUS POLY CONT METAL

Note for Active：

Dark gray and light gray region are both oxide (no difference), different colors used here are just to distinguish different causes of formation.

The dark gray layer (oxide) is to define the Active layer information.

For the dark gray layer, the thin region is Active.

The remain region for SiN after etching is the same as Active.

The final surface morphology of the device is not so flat as the schematic diagram, but will be uneven